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Data-Driven Deep Learning for Imaging Through Scattering Media: Architectures, Learning Paradigms, and Methodological Frontiers

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***Abstract*—Imaging through scattering media continues to be a persistent challenge in optical imaging due to the fact that scattering disrupts the direct relationship between the object and the measured signal. Once this relationship is degraded, reconstructing the original scene turns into a challenging inverse problem, for which standard imaging models are frequently insufficient. Recently, deep learning has emerged as an effective framework for reconstruction, since it allows the mapping from scattered measurements to object estimates to be learned directly from data. This review focuses particularly on purely data-driven methods, in which the neural network acts as the primary reconstruction engine instead of functioning as a supplementary element. The reviewed studies are examined in three dimensions: reconstruction frameworks, learning regimes, and system-level integration. Within this framework, we examine how various approaches trade off reconstruction accuracy, robustness, portability, and computational expense. We also consider training data requirements, adaptation strategies, generalization behavior, and evaluation practice. The literature shows clear progress toward more adaptive and robust reconstruction systems. However, several limitations continue to hinder broader applicability, including dependence on paired training data, fragmented out-of-distribution evaluation, the absence of standardized robustness and benchmarking protocols, and limited reporting on practical deployment. On the basis of this analysis, we identify several priorities for future research, encompassing the development of weakly supervised learning methodologies, the establishment of standardized multi-dimensional robustness assessment protocols, the advancement of modular physics-informed design strategies, and the exploration of more tightly integrated reconstruction frameworks.**

***Index Terms*—Imaging through scattering media; data-driven learning; deep learning; inverse imaging; computational imaging; speckle reconstruction; adaptive reconstruction; generalization.**

I. Introduction

Imaging through scattering media has evolved from a specialized optical challenge into an important inverse imaging problem with implications for biomedical imaging, environmental monitoring, autonomous navigation, and security [1], [2]. In applications such as deep-tissue microscopy, underwater imaging, fog-penetrating vision, and remote sensing, the central difficulty is that scattering disrupts the stable correspondence between the object and the detected signal [1]–[3]. As light propagates through a scattering medium, ballistic photons are increasingly overwhelmed by multiply scattered components, which scramble spatial information and transform the recorded measurement into a complex encoded pattern rather than a conventional image [3]–[5]. As a result, conventional imaging models that rely on direct object-to-image mapping become insufficient for reliable reconstruction through scattering media [1]–[3].

This difficulty has motivated a shift in computational imaging toward learned reconstruction. As illustrated conceptually in Fig. 1, recovering an object from a scrambled optical measurement requires solving an inverse problem that is highly ill-posed, often nonlinear, and sensitive to noise and system variation [1]. Deep learning is attractive in this setting because it can learn a mapping from scattered measurements to object estimates from representative data, thereby reducing the dependence on explicit inversion at inference time [1], [6]. The appeal is clear: a network trained on representative input-output pairs can, in principle, learn to decode the complex speckle patterns into coherent images, enabling rapid, single-pass inference.

From this data-driven perspective, it is essential to clearly state which class of methods is considered in the reviewed studies. We concentrate on pure data-driven reconstruction approaches, which means approaches in which the learned model performs the primary mapping from scattered measurements to the target output [1], [2], [7]. This definition deliberately excludes several related research directions, among them physics-guided inversion, hybrid acquisition, reconstruction frameworks, ghost imaging, and single-pixel imaging—since

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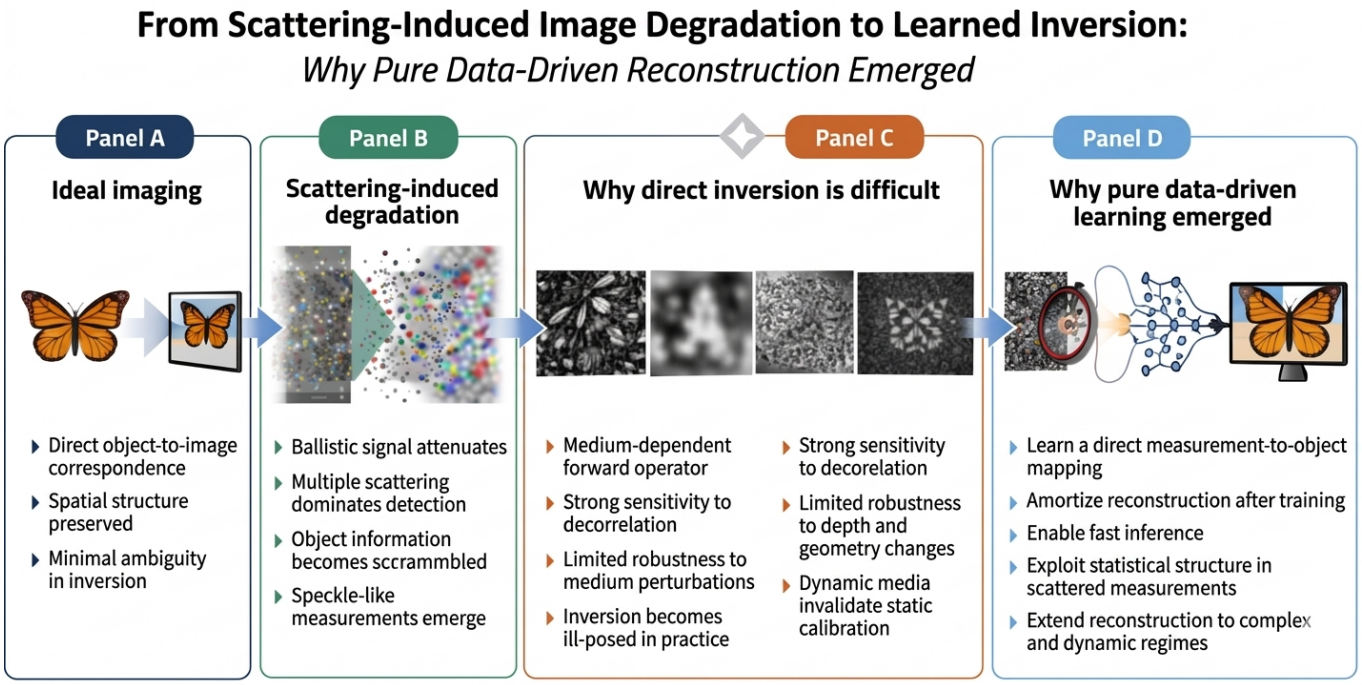


Fig. 1. Conceptual pathway from conventional imaging to pure data-driven reconstruction through scattering media. Panel A represents the direct object-to-image relation in an ideal system; Panel B shows how scattering suppresses ballistic information and produces speckle-like measurements; Panel C summarizes why direct inversion becomes ill-posed under decorrelation, noise, and medium-dependent forward operators; Panel D illustrates the learned measurement-to-object mapping used by pure data-driven reconstruction.

they are based on distinct assumptions and reconstruction principles. Previous review papers have explored related topics, including broader computational imaging and polarimetric imaging in scattering media [1], [2]. The present survey differs from those works in its narrower methodological scope: it focuses on approaches in which the learned model is the primary reconstruction mechanism, and it compares them along architecture, learning paradigm, robustness/generalization evidence, and deployment-oriented trade-offs. We therefore do not position the paper as a PRISMA-style systematic review, but as a structured survey and critical synthesis of a focused methodological subfield. Within this scope, our objective is to assess these methods not only in terms of reconstruction quality, but also with respect to their underlying learning paradigms, documented robustness characteristics, and suitability for real-world deployment.

This review contributes to the existing body of work in four main ways. First, it compiles a well-defined collection of studies focused on purely data-driven deep learning approaches for imaging through scattering media. Second, it provides a comparative analysis of this body of work, emphasizing divergences in model architectures, training paradigms, generalization characteristics, and deployment-related trade-offs. Third, it highlights a set of persistent research-related limitations, including the predominant dependence on paired datasets, heterogeneous evaluation protocols, and the lack of standardized benchmarking frameworks. Fourth, informed by this analysis, it outlines a research agenda aimed at enabling the development of imaging systems that exhibit enhanced robustness, scalability, and real-world applicability.

The rest of the paper is organized as follows. Section II describes the review scope and methodology. Section III presents the foundational context, including the optical characteristics of scattering, the ill-posed nature of the reconstruction problem, and the rise of data-driven approaches. Section IV develops a thematic analysis based on reconstruction backbones, learning regimes, and system-level integration. Section V discusses learning protocols, generalization strategies, and evaluation practices. Section VI summarizes the main research gaps and proposes directions for future work. Section VII concludes with the broader methodological implications of the reviewed literature.

II. Review Scope and Methodology

This structured survey is limited to pure data-driven approaches for imaging through scattering media, where the learned model serves as the main component responsible for reconstructing the target output from scattered measurements. To identify relevant studies, we surveyed peer-reviewed journal and conference publications indexed in Web of Science and Scopus using keyword combinations related to imaging through scattering media, speckle reconstruction, deep learning, self-supervised learning, diffusion models, polarization imaging, and dynamic scattering environments. Greater emphasis was placed on recent publications in order to reflect the current state of methodological development. We included

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studies that directly addressed reconstruction, restoration, or recovery through scattering media using learned models, and we excluded works focused mainly on classical inverse methods, ghost imaging, single-pixel imaging, or physics-guided pipelines in which the learned model did not constitute the primary reconstruction mechanism. For the comparative synthesis, each study was examined across three main dimensions: reconstruction backbone, learning regime, and the extent of physics or hardware integration, together with the reported evidence on data requirements, generalization, robustness, and deployment considerations.

Survey protocol and terminology. This work is a structured survey rather than a PRISMA-style systematic review. Accordingly, the goal is not to claim exhaustive retrieval or meta-analysis, but to synthesize representative peer-reviewed studies that illustrate the main methodological directions in pure data-driven imaging through scattering media. In this paper, *robustness* denotes performance under controlled perturbations of the medium or acquisition setting; *generalization* denotes performance on unseen media, objects, depths, distances, or geometries without complete retraining; *portability* denotes the ability to transfer a method across optical setups with limited recalibration; and *adaptation* denotes explicit model updating, alignment, or fine-tuning during transfer or deployment.

III. Foundational Context: From Optical Scrambling to Learned Inversion

A clear understanding of pure data-driven methods requires attention to the physical and methodological context from which they emerged. This section reviews that foundation through five closely related aspects: the optical characteristics of scattering-induced degradation, the ill-posed nature of the inverse problem, the rise of data-driven inversion as a practical alternative, the range of data representations used in the literature, and the continuing evolution of learning paradigms. Together, these elements provide the conceptual basis for the thematic analysis developed in the following sections.

A. Optical Nature of Scattering-Induced Degradation

Scattering-induced degradation differs fundamentally from conventional blur or additive noise because it disrupts the optical field itself and destroys the direct correspondence between the object and the recorded image [1], [2]. In standard optical imaging, image formation depends mainly on ballistic photons that preserve spatial information from the object to the detector. Once a scattering medium is introduced, however, these ballistic components are progressively masked by multiply scattered light, whose random interference produces a complex speckle-like pattern at the detector [1], [3]. The resulting degradation is therefore not simply a reduction in contrast; rather, it reflects a more fundamental scrambling of information in which the optical field is encoded in a complex and often unknown manner [4], [5]. In effect, the medium converts the object into a measurement that no longer resembles a conventional image. This is the core physical difficulty that pure data-driven methods seek to address by learning the decoding relationship directly from data.

B. Posedness and the Challenge of Inversion

Reconstructing an object from a scattered measurement is fundamentally an ill-posed inverse problem. The relationship between the object domain and the measurement domain is typically non-unique, highly sensitive to small disturbances, and influenced by system parameters that may be unstable or not fully known [1]. The difficulty becomes even more pronounced in dynamic scattering media, where temporal fluctuations modify the effective forward model and may invalidate an inverse mapping learned under earlier conditions [8]–[10]. In addition, the amount of recoverable information can depend on the instantaneous state of the medium, since physical invariants such as ballistic light, when present, may support forms of generalization that purely data-driven models cannot reliably ensure on their own [11]. The problem is further complicated when the reconstruction target extends beyond scalar intensity to quantities such as phase or polarization, for which retrieval is often nonlinear and practically underdetermined [12], [13]. Together, optical scrambling, limited prior knowledge, and medium instability help explain the appeal of data-driven inversion, which offers a practical way to approximate the inverse mapping without relying on a fully tractable physical model.

C. Emergence of Pure Data-Driven Reconstruction

The emergence of pure data-driven reconstruction can be viewed as a practical response to the limitations of explicit inversion methods. Instead of deriving the inverse mapping analytically, which is often intractable, or estimating it through iterative procedures that may be computationally expensive, these approaches learn the relationship directly from representative data [1]. This has been especially appealing in dynamic scattering environments, where conventional techniques often struggle because fluctuations in the medium violate the assumptions behind static reconstruction models [6], [14]. As illustrated in Fig. 2, this paradigm reorganizes the imaging process by connecting optical acquisition, data preparation, and network training to a learned model that can later perform fast, task-specific inference. It is also worth noting that the field did not develop along a single uniform trajectory. What began largely as end-to-end supervised reconstruction has gradually expanded into a broader methodological landscape. While early studies relied heavily on paired supervision, more recent work has moved toward semi-supervised, self-supervised, and label-free or empirical-model-based approaches aimed at reducing data requirements and improving robustness under changing conditions [8], [13], [15]. A defining feature of this paradigm is therefore the shift in computational burden: much of the difficulty of inversion is absorbed during training, allowing inference itself to be comparatively efficient.

D. Data Representations Used in the Reviewed Studies

A crucial yet often overlooked driver of reconstruction quality is the exact form of the optical measurements fed

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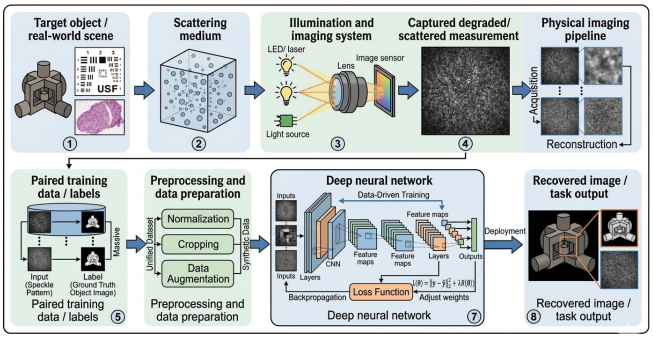


Fig. 2. Workflow of pure data-driven imaging through scattering media. Optical acquisition produces scattered measurements, which are converted into training inputs and paired or weakly paired targets when available; a neural network then learns the measurement-to-object mapping during training and performs fast task-specific reconstruction at inference time. The figure emphasizes that the learned model is the primary reconstruction engine, while preprocessing and acquisition choices determine the information available to the model.

into the network. The studies examined in this review employ a diverse range of input representations. In most cases, the input comprises only lightly processed speckle-intensity patterns, which remain the dominant representation in many direct reconstruction frameworks [16]–[18]. Nevertheless, the suitability and effectiveness of this representation are often dependent on task. For instance, the fusion of complementary inputs—such as demultiplexed speckle patterns and autocorrelation, derived information, can be critical for achieving higher-fidelity reconstructions [6].

A second body of work employs representations that are explicitly informed by physical principles or are optimized for a particular computational objective. Representative examples include the use of diffractograms in ptychographic reconstruction [10], speckle-correlation measurements enhanced via Fourier-domain preprocessing [19], and structured illumination or patterned excitation schemes engineered to reduce the dimensionality of the input data [20]. Collectively, these strategies indicate that advancement in this domain hinges not only on learning from scattered measurements, but also on systematically reformulating the measurement process itself so as to render the associated inverse problem more tractable.

A further line of development involves exploiting multi-dimensional optical information that extends beyond scalar intensity alone. By incorporating vectorial optical fields that jointly encode intensity, phase, and polarization, the network is endowed with additional degrees of freedom that can retain informative features even in the presence of scattering [5]. Among these modalities, polarization has attracted particular interest, as multiple studies have demonstrated that it can provide inputs that are more robust or more discriminative than intensity-only measurements [2], [7], [12], [21]–[23]. Collectively, these advances underscore that data representation should be treated as a primary methodological axis, encompassing a spectrum of choices, from simple raw intensity measurements to physically enriched, information-dense input encodings.

E. Learning Setups and Training Paradigms

The methodological diversity of this field extends beyond reconstruction backbones and input representations to the way the inverse mapping itself is learned. Across the reviewed studies, several learning paradigms appear repeatedly, including fully supervised training with paired data, self-supervised and weakly supervised approaches, transfer learning and adaptation-focused strategies, as well as generative reconstruction frameworks. Fully supervised learning still serves as the dominant baseline and underpins many studies on direct reconstruction and generalization [16]–[18], [22], [24]. At the same time, the field is gradually moving beyond this conventional setting because paired data collection is costly and medium drift can reduce model reliability. This shift is reflected in the increasing use of semi-supervised [8], self-supervised [15], [25], and test-time adaptation methods [3]. Such approaches treat the learning configuration itself as an important design choice for improving scalability and robustness. In parallel, training pipelines are becoming more hybrid, often combining simulated samples, physical modeling, and experimental measurements [19], [20]. Overall, the choice of learning setup, whether based on offline supervision, online adaptation, or domain alignment, is not a secondary detail but a central part of method design that strongly influences the final capabilities of the reconstruction system.

IV. Thematic Analysis Across Reconstruction Backbones, Learning Regimes, and System-Level Integration

Progress in data-driven imaging through scattering media has not been driven by a single dominant model family. Instead, the reviewed studies diversify along three complementary dimensions: reconstruction backbones, learning regimes, and system-level integration. Some studies differ primarily in backbone design, including CNN-, mixer-, transformer-, or diffusion-based reconstruction. Others differ mainly in how learning is performed, including paired supervision, self-supervision, transfer learning, and test-time adaptation. A third group derives its distinctiveness from the coupling between learning and the acquisition system, including physics-coupled and opto-electronic pipelines. Analyzing the literature along these separate axes provides a clearer basis for comparison than treating all of them as architecture families. Table I summarizes this three-axis organization.

Taxonomy boundary. Table I separates three analytical axes: backbone design, learning regime, and degree of system-level integration. These axes are complementary rather than mutually exclusive paper classes. A single study may combine a hybrid CNN/Transformer backbone with an adaptive learning regime and a physics-coupled acquisition design. In this taxonomy, “hybrid CNN/Transformer” refers to network backbone composition, “test-time/online adaptation” refers to the training or deployment update strategy, and “physics-coupled pipeline” refers to explicit coupling between the learned model

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TABLE I
THREE-AXIS TAXONOMY OF PURE DATA-DRIVEN IMAGING THROUGH SCATTERING MEDIA

Axis	Category	Representative Studies	Role in Reconstruction	Strength	Weakness	Computational Cost	Generalization
Backbone	CNN / U-Net variants	[11], [14], [9], [18], [26], [27], [22]	Direct inverse mapping	Accurate; fast; stable; good detail recovery	Performance remains condition-sensitive under strong mismatch	Moderate	Moderate–High
	Transformer / MLP-Mixer	[17]	Global feature modeling	Strong long-range contextual interaction	Higher design and training complexity	Moderate-High	Moderate
	Hybrid CNN/Transformer	[6], [5]	Joint local–global feature modeling	Balanced local detail and global context	Reduced lightweight deployability	High	High
	Diffusion / generative restoration	[3]	Iterative restoration	Strong restoration quality; data-efficient adaptation	Slower inference due to iterative denoising	High	Moderate–High
Learning regime	Paired supervised learning	[16], [17], [18], [24], [22]	End-to-end learning from aligned input–target pairs	Strong baseline performance; stable optimization	High dependence on paired data	Moderate	Moderate
	Semi-/self-supervised learning	[8], [25], [15]	Reducing label dependence and improving robustness	Better scalability; improved adaptability	Higher methodological complexity	Moderate–High	High
	Transfer learning / fine-tuning	[9]	Adapting pretrained models to new scattering conditions	Efficient reuse of pretrained representations	Still depends on domain proximity	Moderate	Moderate–High
	Test-time / online adaptation	[3], [25]	Updating or aligning the model during deployment	Improved resilience to drift and domain shift	Additional runtime and implementation complexity	Moderate–High	High
System integration	Purely digital reconstruction	[11], [14], [3], [26]	Reconstruction performed primarily in the learned digital model	More portable across setups	Less ability to exploit acquisition-side structure	Moderate	Moderate–High
	Physics-coupled pipelines	[10], [19], [20]	Reconstruction integrated with structured acquisition or preprocessing	Strong task-specific performance	Setup-dependent; reduced portability	Moderate–High	Moderate
	Opto-electronic / hybrid optical-digital systems	[16], [24]	Partial optical offloading of reconstruction	Very low digital burden	Optical implementation constraints	Low	Moderate

and structured acquisition, preprocessing, or optical hardware. Thus, overlapping terms are separated according to the design decision they describe.

Within the backbone dimension, CNN- and U-Net-based models continue to serve as the principal baseline in the literature because they support direct inverse reconstruction, can be trained with relatively modest computational overhead, and often deliver strong performance when the training and testing conditions are well matched. Studies such as [9], [11], [14], [18], [22], [26], [27] indicate that these architectures perform particularly well when the scattering environment and acquisition setting remain close to those seen during training. Their effectiveness is largely attributed to their architectural simplicity, their ability to capture local image features, and the stability of supervised optimization. At the same time, their performance is often vulnerable to condition shifts, with noticeable degradation when scattering characteristics, object distributions, or acquisition parameters differ substantially from the training setting.

The literature also points to a gradual shift away from purely local convolutional processing toward models that incorporate wider contextual interactions across speckle patterns. This tendency is illustrated by [17], which employs an MLP-Mixer-based framework for wide-field reconstruction and recognition, emphasizing the importance of long-range feature interaction. A related development can be seen in hybrid CNN/Transformer architectures. For example, [5] combines convolutional and transformer modules to unite local detail extraction with global contextual reasoning, whereas [6] uses transformer-assisted demultiplexing together with U-Net-based reconstruction for dynamic snapshot video recovery. Although these architectures broaden the representational capacity of the reconstruction model, they generally do so at the cost of higher computational demand and reduced suitability for lightweight deployment compared with conventional U-Net-based designs.

Diffusion-based reconstruction represents a separate backbone direction within the reviewed literature. As shown in [3],

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diffusion models can achieve strong restoration performance and remain effective even when the amount of paired training data is limited. Their main limitation, however, is computational cost, since the iterative denoising process typically makes inference slower than in feed-forward reconstruction networks.

Another important source of methodological variation arises from the learning regime rather than from the reconstruction backbone itself. Studies such as [25] and [15] indicate that self-supervised, adaptation-oriented, and physics-guided training strategies can improve robustness under drift and unseen condition changes. These approaches are better understood as differences in training and adaptation strategy, not as distinct architecture families. Their main advantage is greater flexibility in changing environments, whereas their main drawback lies in the additional methodological and implementation complexity they introduce.

A third dimension of variation concerns system-level integration. In physics-coupled pipelines, represented by [10], [19], [20], the effectiveness of reconstruction often depends on combining learning with structured acquisition or preprocessing. Although such systems can perform well when the operating conditions are closely matched to those assumed during development, their portability is often limited because the learned reconstruction remains strongly tied to specific measurement settings. Opto-electronic systems, represented by [16], [24], transfer part of the computational burden to the optical front end and thereby reduce the demand on digital processing. Their main advantage is efficiency, but this benefit is accompanied by tighter implementation and hardware-design constraints.

Overall, the reviewed literature suggests that no single design direction can fully satisfy all major objectives at once, including reconstruction fidelity, robustness, portability, and computational efficiency. In practice, the most suitable method depends on the priority of the application, whether that priority is reconstruction quality, tolerance to condition changes, compatibility with structured acquisition, or deployment efficiency. This diversity reflects both the progress and the complexity of the field, but it also makes clear that more systematic comparative evaluation and clearer design principles are still needed across reconstruction backbones, learning regimes, and system-level integration.

Cross-study implications. The strongest reported generalization evidence is typically associated with approaches that explicitly include domain alignment, self-supervised or test-time adaptation, or physics-guided synthesis [3], [15], [25]. In contrast, fixed paired-supervised feed-forward models are more vulnerable when the scattering medium, optical thickness, object distribution, depth, or acquisition geometry differs substantially from the training condition [9], [11], [26]. From a deployment perspective, feed-forward CNN/U-Net and opto-electronic designs are comparatively attractive in stable operating regimes because they enable single-pass or low-digital-burden inference [14], [16], [24]. However, their reliability depends on the match between training and deployment conditions. Diffusion and hybrid transformer-based models can improve restoration quality or adaptation capacity, but they generally introduce higher computational cost, longer inference, or reduced lightweight portability [3], [5], [6]. Across the reviewed studies, the repeated trade-off is therefore not a single best architecture, but a balance among reconstruction fidelity, OOD robustness, computational cost, setup dependence, and calibration burden.

V. Learning Protocols and Generalization

The methodological development of data-driven imaging through scattering media cannot be understood by considering architecture alone. Equally important are the training protocols used to learn the models, the data regimes on which they depend, the strategies adopted to achieve and assess generalization, and the evaluation practices used to judge robustness and practical utility. As summarized in Table II, the field is undergoing a broader methodological shift. Although reconstruction quality remains a central objective, it is increasingly being examined alongside scalability, reliability under out-of-distribution conditions, adaptability to changing environments, and suitability for real-world deployment.

A. Learning Regimes and Data Frameworks

A primary discernible trend is that the domain largely relies on paired supervised learning. This is apparent in research on scalable descattering, snapshot dynamic reconstruction, diffraction-assisted recovery, adaptive transfer, real-time descattering, and vectorial restoration, where supervised mapping from scattering measurements to target outputs remains the primary training methodology. However, the literature no longer treats strict paired supervision as the only viable paradigm. Study [3] presents adapter-enhanced diffusion in a severely data-constrained environment, study [25] integrates supervised pretraining with online self-supervised modifications, study [9] employs transfer learning and fine-tuning amid varying scattering thickness, and study [15] eliminates the necessity for paired speckle-label acquisition during training via a speckle-free self-supervised approach. Methodologically, while the prevailing regime remains supervised, the field is progressively exploring adaptive and weakly paired alternatives to diminish reliance on comprehensive paired acquisition.

The training data framework elucidates the persistent challenges of scalability in practical applications. Many studies continue to depend on controlled experimental or hybrid datasets that are closely associated with a specific configuration, medium, or acquisition geometry. Study [11] utilizes 20,000 training pairs and assesses generalization on unobserved diffuser regions, while demonstrating that scalability is physically constrained by the persistence of ballistic light across training and testing conditions. Study [3] intentionally minimizes the training requirement to merely 100 paired images from a singular scattering condition, achievable solely through a more robust model prior and test-time adaptation mechanism. Study [25] commences with supervised pretraining utilizing 5,000 to 20,000 paired samples before transi-

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TABLE II
COMPARATIVE ANALYSIS OF LEARNING PROTOCOLS, GENERALIZATION STRATEGIES, AND EVALUATION PRACTICES

Dimension	Representative Studies	Common Practice	Limitation	Impact
Learning regime	[11], [6], [10], [3], [25], [14], [9], [15], [5], [20]	• Mainly paired supervised learning • Selective use of diffusion, transfer learning, online adaptation, and self-supervision	• Strong paired-data reliance • Adaptive/self-supervised schemes increase complexity	• Learning design strongly shapes scalability and robustness
Training data regime	[11], [3], [25], [14], [9], [15], [5]	• Mostly controlled experimental paired datasets • Some studies reduce data demand via small-data training, online updating, or self-supervision	• Costly and setup-specific • Often limited in diversity across media and conditions	• Data regime remains a major bottleneck for real-world transfer
Generalization strategy	[11], [3], [26], [14], [9], [15], [5], [27]	• One-to-all training • Domain alignment and transfer learning • Residual robustness-by-design and self-supervised adaptation	• Severe mismatch and strong scattering change still degrade performance	• Generalization is increasingly treated as a core design objective
OOD / robustness evaluation	[11], [3], [25], [26], [14], [9], [15], [5], [27], [22]	• Increasing use of unseen media, displacement, depth, thickness, distance, drift, material, and angular tests	• Evidence remains fragmented • No unified robustness protocol	• Robustness evidence is growing, but remains uneven

tioning to self-supervised online updates, while study [15] diminishes the necessity for optical paired-data generation by employing physics-guided speckle synthesis and subsequently validating across 440 unseen scattering conditions. These examples substantiate the conclusion presented in Table II: training data are expensive, context-dependent, and frequently lack variation; therefore, the data regime persists as a significant impediment to transfer beyond controlled laboratory settings.

The scale and diversity of training data represent critical factors that remain underexplored. While some studies have demonstrated the feasibility of training with simulated data or small experimental datasets, the systematic relationship between dataset characteristics and reconstruction performance across different scattering conditions has not been established. This represents a significant gap that future research should address through controlled experiments varying dataset size, diversity, and realism.

B. Generalization, Robustness, and Practical Applicability

The recent literature points to a clear shift from reconstruction methods tailored to fixed conditions toward approaches designed with resilience in mind. As summarized in Table II, the strategies explored include one-to-all training, domain alignment, transfer learning, residual robustness, and self-supervised updating. For instance, [11] shows that the feasibility of one-to-all scalability depends on the persistence of ballistic light. Study [3] addresses domain shifts across different media through adapter-based alignment applied to milk, artificial fog, and outdoor fog. Other works target different sources of variation: [26] addresses unseen depth and object variability, [9] adapts to changes in optical thickness, [15] reports generalization across unseen diffusers without paired training, and [5] improves robustness in dynamic biological scattering. Taken together, these studies suggest that generalization is no longer treated as a secondary benefit, but has become a central objective in method design.

At the same time, the available evidence on robustness remains fragmented and largely organized around isolated perturbation axes. As indicated in the table, recent studies increasingly evaluate performance under unseen media, displacement, depth, thickness, distance, drift, material variation, and angular perturbation; however, these factors are rarely examined within a unified evaluation framework. Study [25], for example, provides strong evidence of temporal robustness by reporting over 99.9% transmission across 1 km of multimode fiber for more than 1000 seconds while maintaining video SSIM above 0.995, although the reported setting still involves relatively mild channel drift. Study [3] shows notable gains under cross-domain transfer, yet its performance declines under the most severe fog conditions. Study [15] demonstrates robustness across many unseen variables, but under an evaluation setting that differs substantially from tests of long-term drift or biological dynamics. Likewise, [5] reports robust performance in anisotropic dynamic biological tissue, though within a specific perturbation regime. Overall, the literature points to growing attention to robustness, but the reported evidence remains heterogeneous in both scope and difficulty. Because the reviewed studies use different scattering media, imaging targets, acquisition geometries, training data, and evaluation metrics, this survey does not present a pooled numerical ranking across papers. Instead, reported quantitative values are interpreted within their original experimental context, and the comparison emphasizes recurring patterns in data requirements, robustness evidence, and deployment reporting.

VI. OPEN RESEARCH GAPS AND FUTURE DIRECTIONS

The comparative discussion presented above suggests that data-driven imaging through scattering media has progressed well beyond the stage of proof-of-concept reconstruction. The field now includes direct end-to-end models, adaptive approaches developed to improve robustness, and pipelines that integrate learning with the acquisition process itself. Even so, as shown in Table III, these advances have not yet converged

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into a methodological framework that is broadly applicable across settings. Several core issues remain unresolved, most notably dependence on paired data, limited out-of-distribution (OOD) generalization, fragmented evaluation practices, strong dependence on specific physics-coupled setups, and the continued fragmentation of methodological paradigms.

A. Methodological Gaps

The field continues to face four closely related methodological gaps that limit its progress toward practical real-world use. The first, and arguably the most fundamental, is the continued dependence on controlled paired supervision. Although supervised learning has produced strong results in laboratory settings, its reliance on large, carefully aligned datasets of speckle measurements and corresponding ground-truth targets remains a major obstacle to scalability. Studies such as [3], [15], [25] show that it is possible to move beyond conventional supervision through mechanisms such as diffusion priors, self-supervised updating, and physics-guided synthesis. Even so, dependence on data remains the main constraint on deployment across different media and application scenarios.

A second gap follows directly from this issue, namely limited out-of-distribution (OOD) generalization. Although recent work increasingly emphasizes robustness, most claims of generalization are still tied to particular types of perturbation. A method that performs well on unseen diffusers, for example, may still fail when scattering thickness changes, and a model that tolerates angular misalignment may remain sensitive to changes in material properties. In this sense, robustness is now widely recognized as an important goal, but the field still lacks methods that generalize reliably across the broader range of variability encountered in practice.

The third gap concerns the fragmented way robustness is currently evaluated. Existing studies test different kinds of perturbation, including displacement, temporal drift, medium type, scattering thickness, distance, and angular misalignment, but they do so using different protocols, metrics, and baseline assumptions. As a result, the literature contains many isolated demonstrations of robustness without offering a common basis for determining which approaches generalize most effectively across conditions.

A fourth limitation arises in physics-coupled pipelines, where learning is integrated with coded acquisition, ptychography, or polarization-based measurement design. These methods can achieve impressive performance for specific tasks, but their portability is often reduced because their success depends strongly on particular acquisition assumptions and hardware configurations. This creates a persistent trade-off in the field: methods that perform exceptionally well in specialized settings are often the hardest to transfer across systems.

B. Emerging Research Directions and Guidelines

Addressing these limitations requires moving beyond isolated improvements in model architecture toward reconstruction pipelines that account jointly for supervision, robustness, portability, and deployment constraints. A related requirement is clearer reproducibility-oriented reporting. Future studies should report, when available, code or implementation details, dataset and split construction, scattering-medium parameters, calibration procedures, preprocessing steps, evaluation metrics, computing platform, and inference time or frame rate. When data or code cannot be released, sufficient acquisition or simulation details should be provided to support approximate reproduction and fair comparison.

The first priority concerns learning paradigms. A central objective is to reduce reliance on large paired datasets by incorporating unlabeled measurements, adaptive updating, and physics-informed supervision. Taken together, the studies in [3], [15], [25] suggest that robust reconstruction is possible even with limited paired data when diffusion priors, online self-supervised updates, and physics-guided synthesis are combined in an effective manner. The remaining challenge is to translate these ideas into systematic frameworks that can operate across diverse scenarios without requiring retraining for each individual condition.

The second priority relates to evaluation methodology. Robustness should not be assessed along a single perturbation axis, but across systematically varied conditions, including medium composition, scattering depth, geometric perturbation, temporal drift, and acquisition mismatch, using consistent metrics and matched data splits. The field therefore needs multi-axis OOD evaluation protocols that support meaningful comparison across studies and clarify the actual generalization limits of existing methods. In the absence of such protocols, reported robustness claims remain difficult to compare rigorously.

The third priority concerns the relationship between physics and learning. Physics-coupled approaches often perform well because the measurement process is deliberately structured to make reconstruction easier, as shown in [6], [10], [20]. Yet this advantage often comes at the expense of portability when the learned system is tightly tied to a single acquisition configuration. Future physics-informed approaches should therefore be designed in a modular and recalibratable manner, so that they can adapt across setups without full retraining while still retaining the benefits of structured acquisition.

The fourth, and most integrative, priority is the development of unified reconstruction systems. No existing paradigm optimizes all desired properties at once. Feed-forward models provide computational efficiency, adaptive methods improve resilience under changing conditions, and physics-coupled pipelines exploit structure in the measurement process. Future systems should be designed to combine these strengths from the outset by jointly considering representation choice, learning strategy, adaptation mechanism, physical prior, and deployment requirements. Only through this kind of integrated design can reconstruction systems become simultaneously accurate, robust, portable, and practical to deploy. In practical deployment, reconstruction quality should be considered together with optical hardware complexity, calibration frequency, inference latency, memory and energy demand, acquisition burden,

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TABLE III
OPEN RESEARCH GAPS, FUTURE DIRECTIONS, AND DESIGN GUIDELINES

Gap	Evidence	Impact	Future Direction	Design Guideline
Paired-data dependence	Most pipelines rely on controlled paired supervision; few reduce this burden via diffusion, transfer, or self-supervision.	Limits scalability across media, scenes, and deployment settings.	Expand weakly supervised, self-supervised, and physics-grounded learning.	Avoid exclusive reliance on large paired datasets; incorporate unlabeled data or adaptive updating.
Incomplete OOD generalization	Studies show robustness improvements but still degrade under stronger turbidity, depth change, or material mismatch.	Reduces reliability in real-world heterogeneous environments.	Shift from condition-specific to systematically stress-tested OOD generalization.	Evaluate under hard unseen conditions, not only mild perturbations.
Fragmented robustness evaluation	Studies test different perturbation axes (displacement, drift, medium, thickness, angle); no shared protocol.	Weakens cross-study comparability.	Build standardized multi-axis robustness benchmarks.	Report robustness across several controlled shifts using consistent metrics and split strategy.
High setup dependence	Physics-coupled methods embed learning in coded acquisition, PIE, or polarization encoding.	Strong task performance but reduced portability.	Develop more transferable physics-aware models that tolerate moderate setup variation.	Prefer modular physics integration that can be recalibrated or adapted.
Limited deployment evidence	Only selected studies report latency, FPS, compression ratio, or measurement reduction.	Practical usability hard to assess consistently.	Extend evaluation to include speed, acquisition burden, and system complexity.	Report explicit quality–efficiency trade-offs.
Complexity–portability trade-off	Advanced designs improve fidelity but increase complexity or scenario dependence.	Higher performance in one regime does not guarantee broader usability.	Seek architectures jointly optimizing quality, robustness, and deployment simplicity.	Favor balanced designs with clear benefit for each added complexity.
Paradigm fragmentation	Literature divided among feed-forward, adaptive, and acquisition-coupled pipelines.	Slows convergence toward unified design logic.	Develop unified frameworks combining direct inversion, adaptive robustness, and physical consistency.	Design reconstruction systems as integrated pipelines rather than isolated model classes.

and sensitivity to mechanical or environmental drift.

These directions point to a practical way forward for data-driven imaging through scattering media, moving the field beyond promising but often highly specialized laboratory studies toward approaches that can be reproduced reliably, generalized across conditions, and applied in complex real-world environments.

VII. CONCLUSION

This review considered pure data-driven deep learning methods for imaging through scattering media from three related perspectives: reconstruction architecture, learning strategy, and system-level integration. The literature indicates that no single methodological option can simultaneously maximize reconstruction quality, robustness, efficiency, and portability. CNN- and U-Net-based models remain effective and computationally practical baselines, whereas transformer-based, hybrid, diffusion-driven, adaptive, physics-coupled, and opto-electronic methods offer different strengths depending on the imaging context and the design objectives. At the same time, the field is moving beyond reconstruction methods developed for narrowly constrained settings toward approaches that place stronger emphasis on robustness and adaptability. In particular, self-supervised methods, test-time adaptation, and physics-guided synthesis appear promising for reducing reliance on paired training data and for improving performance under previously unseen conditions. Despite these advances, several important challenges remain. Generalization under out-of-distribution conditions is still limited, evaluation practices are not yet unified, and the tension between high reconstruction performance and broad portability continues to shape methodological choices.

Future work should place greater emphasis on weakly supervised learning, standardized multi-axis robustness benchmarks, modular physics-informed designs, transparent reporting of code and dataset accessibility, and explicit deployment evaluation covering latency, calibration, energy use, and hardware constraints. Progress in these directions will be necessary if the field is to move beyond controlled laboratory demonstrations and toward imaging systems that are reproducible, generalizable, and practical for real-world deployment.

REFERENCES

[1] F. Wang, J. W. Czarske, and G. Situ, "Deep learning for computational imaging: from data-driven to physics-enhanced approaches," *Advanced Photonics*, vol. 7, no. 5, pp. 054 002–054 002, 2025.

[2] X. Li, Y. Han, H. Wang, T. Liu, S.-C. Chen, and H. Hu, "Polarimetric imaging through scattering media: A review," *Frontiers in physics*, vol. 10, p. 815296, 2022.

[3] C. Tang, H. Liu, L. Zhang, Y. Chu, J. Luo, F. Wang, and G. Situ, "Imaging through dynamic scattering media with an adapter-enhanced diffusion model," *Optics Express*, vol. 33, no. 21, pp. 44 775–44 791, 2025.

[4] H. Liang, W. Ye, M. Gao, L. Chen, Y. Liu, J. Wang, J. Li, J. Yuan, and X.-H. Wang, "An optical meta-image-processor for enhanced imaging through strongly scattering media," *Nature Communications*, vol. 16, no. 1, p. 9732, 2025.

[5] Y.-C. Chen, S.-X. Mi, Y.-P. Tian, X.-B. Hu, Q.-Y. Yuan, K.-H. Chew, and R.-P. Chen, "Adaptive vectorial restoration from dynamic speckle patterns through biological scattering media based on deep learning," *Sensors*, vol. 25, no. 6, p. 1803, 2025.

[6] F. Guzmán, E. Vera, and R. Horisaki, "Snapshot video through dynamic scattering medium based on deep learning," *Optics Express*, vol. 33, no. 7, pp. 15 991–16 002, 2025.

[7] K. Yang, F. Liu, S. Liang, M. Xiang, P. Han, J. Liu, X. Dong, Y. Wei, B. Wang, K. Shimizu *et al.*, "Data-driven polarimetric imaging: a review," *Opto-Electronic Science*, vol. 3, no. 2, pp. 230 042–1, 2024.

[8] X. Hu, J. Zhao, J. E. Antonio-Lopez, S. Gausmann, R. A. Correa, and A. Schülzgen, "Adaptive inverse mapping: a model-free semi-supervised learning approach towards robust imaging through dynamic scattering media," *Optics Express*, vol. 31, no. 9, pp. 14 343–14 357, 2023.

Authorized licensed use limited to: SWANSEA UNIVERSITY. Downloaded on September 16,2026 at 18:38:19 UTC from IEEE Xplore. Restrictions apply.

[9] Z. Fu, F. Wang, Z. Tang, Y. Bian, and G. Situ, "Adaptive imaging through dense dynamic scattering media using transfer learning," *Optics Express*, vol. 32, no. 8, pp. 13 688–13 700, 2024.
[10] Y. Liu, G. Hu, X. Chu, Z. Liu, and L. Zhou, "Deep learning based coherent diffraction imaging of dynamic scattering media," *Optics Express*, vol. 31, no. 26, pp. 44 410–44 423, 2023.
[11] X. Zhang, S. Cheng, J. Gao, Y. Gan, C. Song, D. Zhang, S. Zhuang, S. Han, P. Lai, and H. Liu, "Physical origin and boundary of scalable imaging through scattering media: a deep learning-based exploration," *Photonics Research*, vol. 11, no. 6, pp. 1038–1046, 2023.
[12] B. Yang, L. Tan, X. Zhang, T. Zhang, X. Luo, X. Dong, P. Xie, P. Han, and F. Liu, "Learning-based polarization retrieval from intensity speckle of dense scattering media," *Optics Express*, vol. 33, no. 5, pp. 9446–9459, 2025.
[13] H. Tu, H. Liu, T. Pan, W. Xie, Z. Ma, F. Zhang, P. Xu, L. Wu, O. Xu, Y. Xu *et al.*, "Deep empirical neural network for optical phase retrieval over a scattering medium," *Nature Communications*, vol. 16, no. 1, p. 1369, 2025.
[14] H. Liu, F. Wang, Y. Jin, X. Ma, S. Li, Y. Bian, and G. Situ, "Learning-based real-time imaging through dynamic scattering media," *Light: Science & Applications*, vol. 13, no. 1, p. 194, 2024.
[15] Z. Huang, M. Shi, J. Ma, Z. Guo, Y. Gao, and X. Liu, "Speckle-free self-supervised learning for scalable imaging through scattering media with unseen condition changes," *Optics Express*, vol. 32, no. 25, pp. 44 123–44 147, 2024.
[16] Z. Huang, Z. Gu, M. Shi, Y. Gao, and X. Liu, "Op-fcnn: an optronic fully convolutional neural network for imaging through scattering media," *Optics Express*, vol. 32, no. 1, pp. 444–456, 2023.
[17] H. Zhang, L. Wang, Q. Xiao, J. Ma, Y. Zhao, and M. Gong, "Wide-field imaging and recognition through cascaded complex scattering media," *Optics Express*, vol. 32, no. 17, pp. 29 693–29 709, 2024.
[18] H. Lin, C. Huang, Z. He, J. Zeng, F. Chen, C. Yu, Y. Li, Y. Zhang, H. Chen, and J. Pu, "Phase imaging through scattering media using incoherent light source," in *Photonics*, vol. 10, no. 7. MDPI, 2023, p. 792.
[19] B. Wang, Y. Shi, W. Sheng, and Y. Liu, "Imaging through scattering media based on refined speckle-correlation learning," *Optics and Lasers in Engineering*, vol. 195, p. 109263, 2025.
[20] N. Wijethilake, M. Anandakumar, C. Zheng, P. T. So, M. Yildirim, and D. N. Wadduwage, "Deep-squared: deep learning powered de-scattering with excitation patterning," *Light: Science & Applications*, vol. 12, no. 1, p. 228, 2023.
[21] B. Lin, X. Fan, D. Li, and Z. Guo, "High-performance polarization imaging reconstruction in scattering system under natural light conditions with an improved u-net," in *Photonics*, vol. 10, no. 2. MDPI, 2023, p. 204.
[22] B. Lin, X. Fan, and Z. Guo, "Self-attention module in a multi-scale improved u-net (sam-miu-net) motivating high-performance polarization scattering imaging," *Optics Express*, vol. 31, no. 2, pp. 3046–3058, 2023.
[23] X. Ren, Y. Tian, Y. Ren, B. Wang, S. Zhang, A. Hu, K. A. Thakoor, and X. Guo, "Single-shot full-stokes imaging through scattering media," *Optica*, vol. 12, no. 10, pp. 1560–1568, 2025.
[24] Z. Huang, M. Shi, J. Ma, Y. Gao, and X. Liu, "A self-attention armed optronic transformer in imaging through scattering media," *Optics Communications*, vol. 571, p. 130943, 2024.
[25] Z. Li, W. Zhou, Z. Zhou, S. Zhang, J. Shi, C. Shen, J. Zhang, N. Chi, and Q. Dai, "Self-supervised dynamic learning for long-term high-fidelity image transmission through unstabilized diffusive media," *Nature Communications*, vol. 15, no. 1, p. 1498, 2024.
[26] E. Farea, R. A. Saleh, H. Huang, Z. Yan, and J. Han, "Scatresunet: A generalized deep learning model for imaging through variable-depth scattering media," *Journal of Lightwave Technology*, vol. 43, no. 22, pp. 10 228–10 241, 2025.
[27] H. Gao, L. Pei, J. Wang, Z. Hu, B. Bai, T. Ning, K. Hu, and W. Xu, "Enhanced angular misalignment resiliency in multimode fiber imaging leveraging deep learning," *Optics Express*, vol. 33, no. 4, pp. 6697–6708, 2025.

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